

Determinant Definition

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In fact, the determinant is the unique function with these properties! But, proving this with the tools we have is difficult, so we will just take this for granted.

Method to Compute the Determinant

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2. Do operations in reverse following previous rules!

Practice!

Find the determinant of

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

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$$\det(A) = 1$$

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Practice!

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Now you try!

Find the determinant of

$$A = \begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix}$$

Special Types of Matrices

To make our later discussion easier, we define two new kinds of matrices

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Upper triangular

$$U = \begin{bmatrix} u_{11} & u_{12} & \dots & u_{1,n-1} & u_{1n} \\ 0 & u_{22} & \dots & u_{2,n-1} & u_{2n} \\ \vdots & \ddots & \ddots & \vdots & \vdots \\ 0 & 0 & \ddots & u_{n-1,n-1} & u_{n-1,n} \\ 0 & 0 & \dots & 0 & u_{nn} \end{bmatrix}$$

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A matrix $U \in \mathbb{R}^{n \times n}$ is **Upper Triangular** if:

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Definition

A matrix $U \in \mathbb{R}^{n \times n}$ is **Upper Triangular** if:

$$u_{ij} = 0 \text{ for all } 1 \leq j < i \leq n$$

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Definition

A matrix $L \in \mathbb{R}^{n \times n}$ is **Lower Triangular** if:

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$$U = \begin{bmatrix} u_{11} & u_{12} & \dots & u_{1,n-1} & u_{1n} \\ 0 & u_{22} & \dots & u_{2,n-1} & u_{2n} \\ \vdots & \ddots & \ddots & \vdots & \vdots \\ 0 & 0 & \ddots & u_{n-1,n-1} & u_{n-1,n} \\ 0 & 0 & \dots & 0 & u_{nn} \end{bmatrix}$$

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A matrix $L \in \mathbb{R}^{n \times n}$ is **Lower Triangular** if:

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Determinant of a Matrix with a 0 row

A matrix with a 0 row will look something like below

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & 0 & 0 \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

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So,

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We will work with an upper triangular matrix, some good practice would be to do these arguments but for lower triangular matrices!

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We claim that in both cases, the determinant of A is the product of the elements on the diagonal

Less than n pivots

What does this look like?

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This idea extends to larger matrices too! Try to think about what that proof would look like!

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And:

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And:

$$\det(A) = \det(B) \cdot a_{11} \cdot a_{22} \cdot a_{33} = a_{11} a_{22} a_{33}$$

General 2×2 formula

Theorem

Let

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

Then

$$\det(A) = ad - bc$$

$a = 0$ case

Proof.

If $a = 0$, we need to have a pivot in A_{11}

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$$A = \begin{bmatrix} 0 & b \\ c & d \end{bmatrix}$$

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If $a = 0$, we need to have a pivot in A_{11}

$$A = \begin{bmatrix} 0 & b \\ c & d \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{bmatrix} c & d \\ 0 & b \end{bmatrix} = B$$

See that $\det(A) = -\det(B) = -bc$

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See that $\det(A) = -\det(B) = -bc = ad - bc$. □

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If $a \neq 0$, we just need to eliminate c !

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$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \xrightarrow{R_2 = R_2 - \frac{c}{a} R_1} \begin{bmatrix} a & b \\ 0 & d - \frac{bc}{a} \end{bmatrix} = B$$

See that $\det(A) = \det(B) = a \left(d - \frac{bc}{a} \right)$

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If $a \neq 0$, we just need to eliminate c !

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See that $\det(A) = \det(B) = a \left(d - \frac{bc}{a} \right) = ad - bc$. □

General 3×3 formula

We could derive this formula, but it would be easier with Section 4.2, which we will not be covering in class.

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$$A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$$

Then

$$\det(A) = a(ei - fh) - b(di - fg) + c(dh - eg)$$

More Practice

Determine if the determinant of the following systems is 0 or not.

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 4 \\ 1 & 2 & 2 \end{bmatrix}$$

$$B = \begin{bmatrix} 1 & 2 \\ 3 & 1 \end{bmatrix}$$

$$C = \begin{bmatrix} 1 & 2 & 3 & 1 \\ 2 & 4 & 4 & 2 \\ 1 & 2 & 2 & 3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

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We claim that $\det(A^{-1}) = \frac{1}{\det(A)}$.

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